

CSMPRO00000002 V2.0

The Dielectric Citadel Protocol for T.Y. Lin International: Global Bridge Resilience

Version 2.0 | June 2026

Changes V1.0 to V2.0

- AASHTO LRFD 9th Ed 2024: GIC load case now codified in US
- T.Y. Lin portfolio expanded to 400+ bridges
- MXene SE=92 dB replaces V1.0 graphene paint SE=15 dB
- BFRP/Elum(r) recyclable; FHWA EM-Resilience funding eligible
- Solar Cycle 25 peak SSN ~137: 12% higher GIC field than V1.0 design basis

1. T.Y. Lin Portfolio GIC Vulnerability Tiers

Tier 1 (critical — immediate retrofit): - Golden Gate approaches + Bay Bridge SAS (SF) - Tilikum Crossing (Portland) - Gordie Howe International Bridge (Windsor-Detroit, under construction 2025)

Tier 2 (2-5 year retrofit): All cable-stayed/suspension bridges > 500m span **Tier 3 (5-10 year):** Concrete girder and arch bridges (BFRP rebar risk)

2. Aegis Bridge Package V2.0 for T.Y. Lin

Package A — Cable Retrofit: - ZrO₂ ceramic cable band insulators every 100m: \$4,500/band - MXene Ti3C₂Tx spray on cables: \$28/m² - BFRP replacement stay cable pilot: \$850/m

Package B — Deck Retrofit: - YInMn Blue QD-enhanced deck coating: \$8/m² - BFRP rebar overlay: \$45/m²

Package C — Monitoring: - FBG optical strain network: \$3,200/bridge - LoRa mesh bridge health (GPS-free): \$8,000/bridge

Total T.Y. Lin Tier 1 program (~48 bridges x \$2.4M avg): \$115M FHWA EM-Resilience eligible (80% match): CSM revenue potential \$23M

End of CSMPRO00000002 V2.0 | Carrington Storm Motors